EXPLORATORY DATA ANALYSIS

CHAPTER 9

Distribution Shape: Skewness, Kurtosis, and Quantiles

Measuring asymmetry, tail behavior, and relative position—without mistaking a statistic for the distribution itself

Focus: moment and quantile skewness, kurtosis conventions, robust shape summaries, quantile interpretation, and the limitations of shape coefficients.

Scope: descriptive distribution analysis. This chapter does not cover inferential tests of normality, transformations, or model assumptions.

Learning Objectives

- Interpret skewness as asymmetry and distinguish visual asymmetry from a numerical coefficient.
- Understand the third standardized moment and why sample skewness formulas differ.
- Explain kurtosis, excess kurtosis, and the relationship between kurtosis and tail behavior.
- Use quantiles, percentile ratios, and quantile-based shape measures to describe distributions.
- Recognize how outliers, sample size, ties, and bounded data affect shape statistics.
- Calculate shape summaries with Pandas and SciPy and state conventions clearly.
- Compare distribution shape across groups without overinterpreting small differences.

Review of Earlier Chapters

Chapter 1 — What Is Exploratory Data Analysis?

EDA is iterative investigation. Quantitative summaries and visualizations complement one another; neither automatically explains the data-generating process.

Chapter 2 — Understanding the Dataset

Variable meaning, observational unit, measurement scale, granularity, and context determine what a distribution and its shape actually represent.

Chapter 3 — Data Quality Assessment

Invalid values, duplicates, missingness, inconsistent units, rounding, and collection limits can create artificial skew, spikes, gaps, or tails.

Chapter 4 — Exploring Numerical Variables

Mean, median, spread, quartiles, IQR, and outlier investigation provide the foundations for understanding numerical distributions.

Chapter 5 — Exploring Categorical Variables

Frequency tables and category proportions describe categorical distributions. Shape coefficients such as skewness are not generally meaningful for nominal labels.

Chapter 6 — Exploring Date and Time Variables

Durations, waiting times, and event gaps often have asymmetric distributions. Their shape must be interpreted in the context of schedules, exposure, and event-generation processes.

Chapter 7 — Descriptive Statistics: Beyond the Basics

Robust summaries, quantiles, weighted means, and SD/SE distinctions help describe data beyond a single average.

Chapter 8 — Understanding Distributions

Histograms, ECDFs, density estimates, modality, gaps, and tails show the empirical distribution. Bin width and KDE bandwidth can change visual impressions.

Review of Chapter 8 — Understanding Distributions

- Frequency counts observations; relative frequency divides counts by valid n; density rescales histogram height so area is one.
- Histogram appearance depends on bin edges and width; compare multiple reasonable choices.
- An ECDF shows the proportion at or below each threshold without binning.
- KDE is a smoothed estimate whose appearance depends on bandwidth; use caution for discrete and bounded variables.
- Compare groups with common scales, bins, and observation windows.
- Investigate spikes, gaps, zero inflation, and apparent multiple peaks before assigning explanations.

Chapter 8 Assignment and Project — Discussion Guide

No assignment submission was included, so this is a reasoning guide rather than a review of your specific answers.

- For x = [0, 0, 1, 1, 2, 2, 2, 3, 4, 5, 8, 13, 21, 34], the distribution is discrete, nonnegative, and has a long upper tail; the large values are increasingly sparse.
- A frequency table is appropriate because the sample contains repeated discrete values. Relative frequencies should use the valid observation count as denominator.
- An ECDF is especially useful for answering threshold questions, such as the proportion at or below 5.
- Different histogram bins can alter apparent peaks; conclusions about gaps and tails should also be checked against the raw values and ECDF.
- A group-comparison project should use consistent bins and axes and report group sizes and missingness.

Expected project learning: inspect complete distributions using complementary views, understand visualization parameters, and distinguish empirical patterns from possible explanations.

Preview of This Chapter

This chapter adds numerical descriptions of distribution shape. The goal is not to label a dataset with one word, but to quantify asymmetry, tail behavior, and relative positions while understanding the assumptions and limitations behind each measure.

1. Shape Is More Than One Number

A distribution's shape includes symmetry, skewness, tail behavior, modality, concentration, and gaps. Skewness and kurtosis are useful descriptors, but they do not uniquely identify a distribution.

Feature	Question it helps answer
Skewness	Is one side of the distribution more extended or influential?
Kurtosis	How prominent are extreme deviations relative to the variance scale?
Quantiles	Where do specified proportions of observations lie?
Quantile ratios	How does spread or tail distance compare across the distribution?
Visual inspection	Are there clusters, gaps, bounds, or unusual structures?

2. Skewness: Concept and Interpretation

Skewness describes asymmetry. A right-skewed distribution often has a longer or more influential upper tail; a left-skewed distribution often has a longer or more influential lower tail. The sign indicates direction, not the practical severity or cause.

- Positive skewness: asymmetry influenced toward the upper/right tail.
- Negative skewness: asymmetry influenced toward the lower/left tail.
- Skewness near zero: no strong third-moment asymmetry; this does not prove the distribution is symmetric.

The mean is often pulled toward a long tail, so mean greater than median can be a clue to right skew and mean less than median a clue to left skew. These are heuristics, not definitions or proofs.

3. Moment-Based Skewness

For a population with finite, nonzero standard deviation, the standardized third central moment is:

$$\gamma_1 = E[(X - \mu)^3] / \sigma^3$$

Cubing deviations preserves their sign and gives greater influence to observations far from the mean. A few extreme values can therefore strongly affect moment-based skewness.

For a sample, software packages use different finite-sample adjustments. Pandas Series.skew() uses an adjusted Fisher–Pearson sample skewness estimator and excludes missing values by default. Other formulas may produce different answers, especially for small samples.

```
x = pd.Series([2, 3, 3, 4, 5, 6, 8, 20])

pandas_skew = x.skew()
scipy_skew_bias_corrected = stats.skew(
    x.dropna(), bias=False
)
scipy_skew_unadjusted = stats.skew(
    x.dropna(), bias=True
)
```

When reproducibility matters, report the library, function, and bias setting. Do not compare coefficients computed with different conventions as if they were identical.

4. Quantile-Based Skewness

Quantile-based measures describe asymmetry through positions rather than cubed deviations. One common measure is Bowley's (quartile) skewness:

B = (Q3 + Q1 − 2Q2) / (Q3 − Q1)

Here Q2 is the median. The measure compares the distance from the median to the upper quartile with the distance from the lower quartile to the median. Its value is bounded between –1 and +1 when Q3 is greater than Q1.

```
q1, q2, q3 = x.quantile([.25, .50, .75])
iqr = q3 - q1

bowley_skew = (
    (q3 + q1 - 2 * q2) / iqr
    if iqr != 0 else np.nan
)
```

Bowley's measure is less sensitive to extreme endpoints than moment skewness, but it only uses the central half of the distribution. It may miss asymmetry concentrated in the far tails. If IQR is zero, the measure is undefined.

5. Kurtosis: What It Does and Does Not Mean

Kurtosis is based on the standardized fourth central moment. For a population with finite, nonzero variance:

$\beta_4 = E[(X - \mu)^4] / \sigma^4$

Because deviations are raised to the fourth power, observations far from the mean have substantial influence. Kurtosis is therefore sensitive to tail observations.

Excess kurtosis subtracts 3, the kurtosis of a normal distribution:

$\gamma_4 = \beta_4 - 3$

Convention	Normal distribution reference	Interpretation
Pearson kurtosis β_4	3	Fourth standardized moment
Excess kurtosis γ_4	0	Pearson kurtosis minus 3

Positive excess kurtosis is often associated with more extreme tail observations relative to a normal reference; negative excess kurtosis is often associated with lighter tails. However, kurtosis is not simply 'peak height'. A coefficient alone does not reveal whether a distribution is unimodal, sharply peaked, or has multiple clusters.

6. Sample Kurtosis and Software Conventions

Sample kurtosis estimators vary in bias correction and in whether they return Pearson kurtosis or excess kurtosis. Pandas Series.kurt() returns unbiased sample excess kurtosis (normal reference 0). SciPy's stats.kurtosis defaults to Fisher=True and bias=True; set bias=False for a bias-corrected estimate.

```
values = x.dropna()

pandas_excess = values.kurt()
scipy_excess = stats.kurtosis(
    values, fisher=True, bias=False
)
scipy_pearson = stats.kurtosis(
    values, fisher=False, bias=False
)
```

A normal-reference value of zero refers to excess kurtosis, not Pearson kurtosis. Always state which convention is being reported. Estimates can be very unstable for small samples and highly sensitive to extremes.

7. Quantiles: Position Within the Distribution

The p-quantile is a value q such that approximately proportion p of observations lie at or below q. The median is the 0.50 quantile; the 0.90 quantile is the value at or below which approximately 90% of observations fall.

```
quantiles = x.quantile([
    .01, .05, .10, .25, .50,
    .75, .90, .95, .99
])
```

Quantiles answer practical questions: What value marks the top 10%? What is the 95th percentile of response time? What range contains the middle 80%? For discrete data and small samples, quantiles may be interpolated and need not be observed values.

Pandas defaults to linear interpolation. Other conventions—lower, higher, nearest, and midpoint—can return different values. Record the method when exact reproducibility or comparison is important.

8. Quantile Spreads and Tail Measures

Quantile differences describe spread in selected parts of a distribution.

Measure	Formula	What it describes
IQR	Q3 – Q1	Width of middle 50%
P90 – P10	Q0.90 – Q0.10	Width of central 80%
P95 – P05	Q0.95 – Q0.05	Width of central 90%
Upper-tail gap	P99 – P50	Distance from median to upper 1% threshold
Lower-tail gap	P50 – P01	Distance from lower 1% threshold to median

These are descriptive tail-distance measures, not formal tail-index estimators. They are often easier to explain to stakeholders than kurtosis, but extreme quantiles require sufficient sample size.

9. Quantile Ratios and Relative Tail Asymmetry

Ratios can compare spread across scales when the denominator is positive and meaningful. For strictly positive variables, P90/P50 or P50/P10 can describe relative upper and lower spread. Ratios become unstable or meaningless when denominators are zero, negative, or close to zero.

```
q = x.quantile([.10, .50, .90, .99])

upper_ratio = (
    q.loc[.90] / q.loc[.50]
    if q.loc[.50] > 0 else np.nan
)
```

A difference such as P90 – P50 is often safer than a ratio. Choose the form that answers the domain question and explain the units or dimensionless interpretation.

10. Robustness, Outliers, and Sample Size

Moment-based skewness and kurtosis are sensitive to extreme values. Quantile-based measures are generally more robust, but high quantiles such as P99 still depend on a relatively small number of observations.

- With small n, the upper and lower 1% quantiles are not reliably resolved.
- With many ties or discrete values, IQR can be zero even when the variable is not constant.
- Measurement limits, censoring, rounding, and truncation can change apparent shape.
- One extreme observation can materially alter skewness and kurtosis; investigate it rather than automatically deleting it.
- Report sample size and missingness alongside shape coefficients.

11. A Practical Shape Profile in Python

The following function reports complementary moment and quantile measures. It labels sample skewness and excess kurtosis explicitly.

```
def shape_profile(series):
    x = pd.to_numeric(series, errors="coerce")
    v = x.dropna()

    if v.empty:
        return {
            "valid_n": 0,
            "missing_n": int(x.isna().sum())
        }

    q = v.quantile([
        .01, .05, .10, .25, .50,
        .75, .90, .95, .99
    ])
    iqr = q.loc[.75] - q.loc[.25]

    return {
        "valid_n": len(v),
        "missing_n": int(x.isna().sum()),
        "sample_skewness": v.skew(),
        "sample_excess_kurtosis": v.kurt(),
        "bowley_skewness": (
            (q.loc[.75] + q.loc[.25]
             - 2 * q.loc[.50]) / iqr
            if iqr != 0 else np.nan
        ),
        "q01": q.loc[.01],
        "q05": q.loc[.05],
```

```
        "q10": q.loc[.10],
        "q25": q.loc[.25],
        "median": q.loc[.50],
        "q75": q.loc[.75],
        "q90": q.loc[.90],
        "q95": q.loc[.95],
        "q99": q.loc[.99],
        "iqr": iqr,
        "p90_p10": q.loc[.90] - q.loc[.10]
    }
```

This function is a starting point, not a universal diagnostic. Add units, variable-specific validity checks, and warnings for small samples or degenerate distributions.

12. Visual Diagnostics for Shape

Use the histogram and ECDF together. A histogram highlights local concentration and modality; an ECDF gives a stable cumulative view and makes quantile positions easier to inspect.

```
fig, axes = plt.subplots(1, 2, figsize=(10, 4))

axes[0].hist(
    x.dropna(), bins="fd",
    edgecolor="black"
)
axes[0].set_title("Histogram")
axes[0].set_xlabel("Value")
axes[0].set_ylabel("Frequency")

sorted_x = np.sort(x.dropna().to_numpy())
ecdf = np.arange(1, len(sorted_x) + 1) / len(sorted_x)
axes[1].step(sorted_x, ecdf, where="post")
axes[1].set_title("ECDF")
axes[1].set_xlabel("Value")
axes[1].set_ylabel("Cumulative proportion")

fig.tight_layout()
```

A Q–Q plot can compare sample quantiles with theoretical quantiles, but formal normality assessment and inferential testing are outside this chapter. Do not use a single Q–Q plot or coefficient as a substitute for domain reasoning.

13. Comparing Shape Across Groups

Group-wise skewness and kurtosis can be useful, but coefficients from small groups are noisy. Compare sample sizes, quantiles, histograms, and ECDFs alongside the coefficients.

```
group_shape = (
    df.groupby("segment")["amount"]
      .apply(lambda s: pd.Series(shape_profile(s)))
      .unstack()
)
```

For clear reporting, create a group summary table with n, median, IQR, P10, P90, skewness, and excess kurtosis. Investigate whether group membership, time period, or measurement practices differ.

14. Common Mistakes

- Treating skewness near zero as proof of symmetry.
- Assuming the mean–median relationship defines skewness.
- Calling kurtosis simply peak height or interpreting it without its convention.
- Comparing Pearson kurtosis with excess kurtosis as if they were the same scale.
- Ignoring small sample size and estimator bias settings.
- Reporting P99 for a tiny sample as if it were a stable tail estimate.
- Using quantile ratios with zero or negative denominators.
- Treating quantile interpolation differences as data errors.
- Deleting extremes just to make shape coefficients look more familiar.
- Reporting shape statistics without units, valid n, or visual context.

Chapter Summary

- Skewness describes asymmetry; moment skewness is sensitive to extreme deviations.
- Bowley's quartile skewness uses Q1, median, and Q3 and is more robust to extremes.

- Kurtosis is based on the standardized fourth central moment; excess kurtosis subtracts 3.
- Positive excess kurtosis is associated with relatively influential tail observations, but kurtosis is not merely peak height.
- Quantiles locate observations within a distribution; quantile spreads describe selected regions.
- High quantiles require adequate sample sizes and careful interpretation.
- State software conventions for skewness, kurtosis, and quantile interpolation.
- Combine coefficients with histograms, ECDFs, group context, and data-quality checks.

Assignment

Part A — Conceptual Questions

- What does positive skewness indicate? What does it not tell you?
- Why can a distribution with skewness near zero still be asymmetric?
- Explain the third standardized central moment.
- Distinguish Pearson kurtosis from excess kurtosis.
- Why is kurtosis not simply a measure of peak height?
- Compare moment-based and Bowley skewness.
- Why can P99 be unreliable for a small dataset?
- Explain why quantile interpolation conventions matter.
- When are quantile ratios inappropriate?
- Why should skewness and kurtosis always be interpreted with visual evidence?

Part B — Manual and Computational Analysis

```
x = pd.Series([
    2, 3, 3, 4, 4, 5, 5, 6,
    7, 8, 9, 12, 18, 35
])
```

- Calculate Q1, median, Q3, IQR, P10, P90, P95, and P99.
- Calculate Pandas sample skewness and excess kurtosis.
- Calculate Bowley's quartile skewness.
- Compare mean and median and explain what they suggest about asymmetry.
- Plot a histogram and ECDF. Explain which features each makes easier to see.
- Discuss how the largest value affects moment-based shape measures.

Part C — Group Comparison

- Choose a dataset with at least two groups and one numerical variable.
- Report valid n, median, IQR, P10, P90, skewness, and excess kurtosis per group.
- Compare group histograms and ECDFs using common axes.
- Identify which shape differences appear substantial and which may be sample-size artifacts.
- Write two cautious interpretations and list alternative explanations.

Part D — Coding Assignment

- Implement shape_profile() and document the formulas and software conventions.
- Add a configurable quantile list and interpolation method.
- Add Bowley skewness with safe handling of zero IQR.
- Compare Pandas and SciPy skewness/kurtosis outputs and explain differences.
- Test on symmetric, right-skewed, left-skewed, heavy-tailed, bounded, and small-sample data.
- Include plots and a short interpretation that avoids causal claims.

Coding Project — Distribution Shape Analysis Module

Extend the cumulative EDA master notebook with a shape-analysis module that builds on Chapters 7 and 8. Keep the functions reusable and avoid duplicating earlier summary code.

- Implement shape_profile() with n, missingness, moment skewness, excess kurtosis, quantiles, IQR, and selected quantile spreads.
- Implement Bowley's quartile skewness and clearly handle zero IQR.
- Allow users to specify quantile probabilities and interpolation convention.
- Compare Pandas and SciPy estimators with documented bias and Fisher/Pearson settings.
- Create group-wise shape summaries with sample-size warnings.
- Generate histogram and ECDF diagnostics using Matplotlib's object-oriented API.
- Test on distributions with different symmetry, tails, bounds, ties, and sample sizes.
- Write a concise report explaining what the coefficients reveal and what they cannot establish.

Expected reusable components: shape_profile(), bowley_skewness(), and grouped_shape_summary(). Include docstrings, input validation, explicit conventions, and examples.

Deliverables: updated cumulative EDA master notebook, group comparison table, diagnostic plots, and a short written interpretation.

Preview of Next Chapter

Chapter 10 — Visualizing Distributions

We'll move from interpreting distribution shape to designing effective distribution visualizations: histograms, KDEs, ECDFs, boxplots, violin plots, strip and swarm plots, rug plots, and small multiples. We'll focus on choosing the right visual form, controlling misleading defaults, and communicating distribution structure clearly.